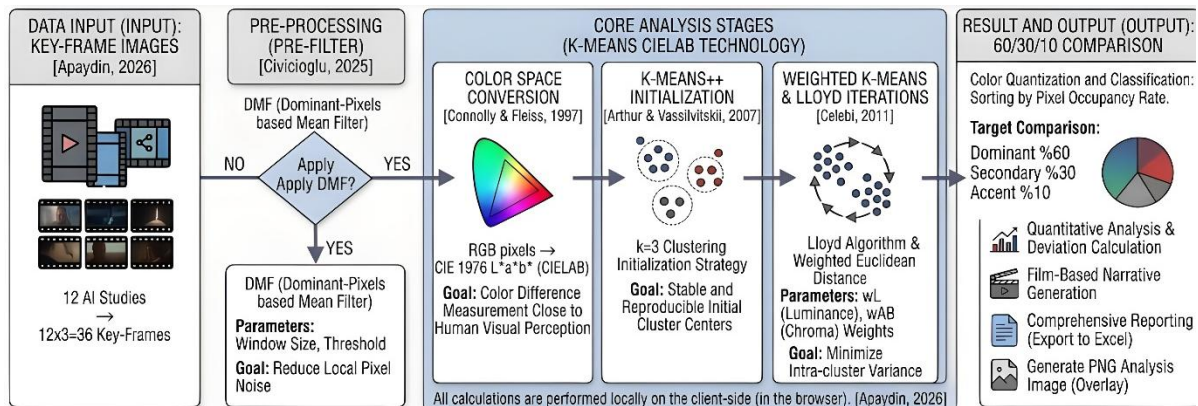


CIELAB-Based K-Means Color Palette Extraction



In this study, the CIELAB-Based K-Means Color Palette Extraction method was used to determine the dominant colors in keyframes in a data-driven and reproducible manner. First, the image is reduced to maintain the speed of analysis. When selected by the user, small-scale pixel noise is reduced by applying a Dominant-Pixels-based Mean Filter (DMF). Then, the non-transparent pixels are taken according to the selected sampling density. These pixels are converted from RGB space to CIE 1976 space. The CIELAB space supports the Euclidean distance between colors to represent the visual color difference more meaningfully than the RGB channel values (Connolly & Fleiss, 1997). $L^*a^*b^*$

Cluster centers are selected using the K-Means++ initial method. This approach reduces the instability that can be caused by random start by allowing the starting centers to be located farther apart from each other in the data space (Arthur & Vassilvitskii, 2007). Each pixel is then assigned to the nearest center; the center of each cluster is updated with the average value of the pixels assigned to it. The assignment and update steps are continued until the assignments do not change or a maximum of 25 iterations are reached. Thus, a representative color palette is produced that reduces the color difference within the cluster. The fact that K-Means is an effective approach to color quantization with appropriate initialization strategies has been discussed in detail by Celebi (2011). $L^*a^*b^*$

Two representation formats are presented for each cluster. The "Average" option converts the cluster center to RGB and gives the theoretical average color. The "Real Pixel" option reports the source image pixel with the lowest CIELAB distance to the cluster center. This option ensures that the reported color is actually present in the image and is associated with the medoid representation (Park & Jun, 2009). Clusters are ranked from largest to smallest according to their pixel occupancy rate; the first three clusters are assigned as dominant, secondary, and accent colors. In the final stage, these ratios are compared to the 60 percent, 30 percent, and 10 percent targets. The method does not prioritize color families; it derives the palette directly from the color distribution in the image data.

APA 7 References

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